

Perturbing the Visual Field: A Spatiotemporal Hypothesis of Consciousness

George Wagenknecht

Abstract

Consciousness is commonly treated as a phenomenon generated by neural activity within the human brain. This paper considers an alternative hypothesis: that conscious experience is not an exclusively brain-local phenomenon, but arises through the interaction between neural processing and the broader spatiotemporal structure in which the brain is embedded. Visual perturbation provides a practical experimental probe because controlled changes in the visual field can alter perceptual processing, attention, decision-making, reaction time, and measurable cognitive effort. The proposed framework asks whether such effects are completely explained by brain-local variables or whether systematic residual dependencies remain associated with the temporal and spatial structure of the organism's environment. The hypothesis does not require a supernatural mechanism; it instead treats cognition as a dynamical process occurring through continuous brain-body-environment interaction.

1. Introduction

The conventional scientific picture places consciousness within the activity of the nervous system. Visual information enters the sensory system, undergoes increasingly complex processing, and contributes to perception, attention, memory, decision-making, and conscious report. This framework has considerable explanatory power. However, a deeper question remains concerning the relationship between the observer and the physical world.

A human brain does not operate in isolation. It continuously receives structured information from its environment. Photons arrive at the retina according to a spatial and temporal distribution; neural activity responds to these signals; muscles alter the body's position; and those movements change subsequent sensory input. The resulting system is therefore inherently dynamical.

2. The Central Hypothesis

The central hypothesis is that the temporal-spatial structure of the physical environment may participate in the dynamics conventionally identified as consciousness. This can be expressed as:

$$C(t) = F[B(t), X(t), T(t), S(t)]$$

where $B(t)$ represents neural state, $X(t)$ sensory interaction, $T(t)$ temporal structure, and $S(t)$ spatial structure. This does not require consciousness to exist outside the brain. Rather, it proposes that the relevant causal system may be larger than the brain alone: brain + body + environment.

3. Visual Perturbation as an Experimental Probe

Vision provides an unusually useful experimental system because the visual field can be perturbed with high temporal and spatial precision. Let the visual stimulus be $V(x,y,t)$. A perturbation can be represented as:

$$V'(x,y,t) = V(x,y,t) + P(x,y,t)$$

where P represents a controlled perturbation. Examples include temporal flicker, spatial displacement, contrast modulation, phase reversal, apparent-motion patterns, masking, controlled visual noise, and changes in spatial frequency or temporal predictability.

4. Measuring Cognitive Effort

The participant can perform a controlled cognitive task while behavioral, oculomotor, neural, and physiological measurements are collected. Candidate observables include reaction time, accuracy, pupil diameter, fixation duration, EEG or MEG measures, heart-rate variability, and skin conductance. A composite effort index may be defined for exploratory analysis, although multivariate modeling is preferable to an arbitrary single score.

$$E_c = w_1 RT + w_2 ER + w_3 D + w_4 P + w_5 N$$

5. The Critical Experimental Comparison

The decisive comparison is between stimuli that are matched on conventional measures while differing in temporal or spatial organization. If two stimuli have approximately equal information content but different temporal structure, then a difference in cognitive dynamics would establish temporal organization as an important explanatory variable. An analogous manipulation can isolate spatial structure.

$$H(V_1) \approx H(V_2), \quad T(V_1) \neq T(V_2)$$

6. Predictive Residuals

A stronger test first constructs a conventional brain-based predictive model of cognitive effort. Let $\bar{E}_{\text{brain}} = F(B)$ denote the effort predicted from measured neural variables. The unexplained residual is:

$$R(t) = E_c(t) - \bar{E}_{\text{brain}}(t)$$

The critical question is whether $R(t)$ contains systematic information about externally controlled spatiotemporal variables after controlling for sensory input, attention, task difficulty, neural state, and physiology. A reliable relationship would not by itself prove that consciousness exists outside the brain; it would instead show that a brain-local explanatory model is incomplete.

7. The Continuum Hypothesis

The organism can be represented as a feedback system in which environmental state influences neural state, neural state influences action, and action changes the subsequent environment:

$$E_t \rightarrow B_t \rightarrow A_t \rightarrow E_{t+1}$$

Under this interpretation, consciousness may correspond not to a particular object inside the loop but to a dynamically sustained process across it. The analogy is similar to a whirlpool: the phenomenon is physically real, but it is not identical to any individual molecule of water.

8. Is Consciousness 'Fake'?

The word 'fake' requires a distinction. If consciousness is defined as a separate inner substance or independently existing entity, a physical account could make that conception unnecessary. But this does not imply that experience or information processing is unreal. A useful distinction is consciousness as substance versus consciousness as process. The latter can be physically real while the intuitive concept of an independent inner observer is partly constructed.

9. Falsifiable Predictions

The framework makes several testable predictions: temporal perturbations should affect cognitive effort even when average visual information is controlled; spatial perturbations should produce effects beyond simple intensity changes; neural activity should exhibit measurable coupling to external temporal structure; and, if a brain-only model is sufficient, externally controlled variables should cease to explain residual variance once the model is adequately specified.

10. Alternative Explanations

Observed effects could arise from altered sensory processing, redistribution of attention, increased prediction error, working-memory demands, arousal, neural oscillatory synchronization, or changes in task strategy. Therefore, visual perturbation

alone cannot establish environmental consciousness. The appropriate scientific claim is narrower: the experiment tests whether cognitive dynamics can be completely accounted for by a brain-local model under controlled spatiotemporal perturbation.

11. Discussion

The proposed framework shifts the consciousness problem toward dynamical-systems theory. Rather than treating consciousness as an object hidden somewhere inside the brain, it treats cognition as a process occurring through continuous interaction. This does not deny neuroscience; neural activity remains indispensable. The proposal is that neural activity may be one component of a larger causal architecture.

12. Conclusion

The apparent boundary between 'inside the brain' and 'outside the brain' may be less fundamental for cognition than ordinary intuition suggests. A human observer is continuously embedded within a temporally and spatially structured physical world. Visual perturbation experiments provide a practical way to investigate this relationship.

The key scientific question is therefore not simply whether the brain generates consciousness, but whether the dynamics identified as consciousness are completely determined by brain-local variables or depend irreducibly upon the organism's temporally and spatially structured interaction with its environment. If the former is demonstrated, the brain-only model gains support. If systematic residual dependencies remain after rigorous control, consciousness may need to be described at the level of an extended dynamical system. In that interpretation, consciousness is not a mysterious substance but a process occurring through physical continuity.

Author: George Wagenknecht

Note: This document presents a speculative, falsifiable research hypothesis and proposed experimental framework; it should not be read as establishing that consciousness is literally non-local or independent of the brain.